1. The equation of an ellipse whose focus is (-1, 1), whose directrix is $x-y+3=0$ and whose eccentricity is $\frac{1}{2}$, is given by

(a) $7x^2+2xy+7y^2+10x-10y+7=0$

(b) $7x^2-2xy+7y^2-10x+10y+7=0$

(c) $7x^2-2xy+7y^2-10x-10y-7=0$

(d) $7x^2-2xy+7y^2+10x+10y-7=0$

2. The locus of the point of intersection of the perpendicular tangents to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ is

(a) $x^2+y^2=9$ (b) $x^2+y^2=4$

(c) $x^2+y^2=13$ (d) $x^2+y^2=5$

3. If a circle and a parabola intersect in 4 points then the algebraic sum of the ordinates is

(a) Proportional to the arithmetic mean of the radius and the latus rectum

(b) Equal to zero

(c) Equal to the ratio of the arithmetic mean and the latus-rectum

(d) None of these

4. The slopes of two tangents drawn from (1, 4) to the parabola $y^2 = 12x$ are

(a) 1, 4 (b) 1, 3 (c) 1, 2 (d) 2, 3

5. Three normals to the parabola $y^2 = x$ are drawn through a point (c, 0) then

(a) c =1/4 (b) c=1/2 (c) c > ½ (d) c<1/2

6. Curve $16x^2 + 8xy + y^2 - 74x - 78y + 212 = 0$ represents

(a) Parabola (b) Hyperbola

(c) Ellipse (d) None of these

7. The length of the latus-rectum of the parabola whose focus is $\left(\frac{u^2}{2g}\sin 2\alpha, -\frac{u^2}{2g}\cos 2\alpha\right)$ and directrix is y = $\frac{u^2}{2g}$ is

(a) $\frac{u^2}{g}\cos^2\alpha$ (b) $\frac{u^2}{g}\cos 2\alpha$

(c) $\frac{2u^2}{g}\cos^2 2\alpha$ (d) $\frac{2u^2}{g}\cos^2\alpha$

8. From any point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}$ = 1, tangents are drawn to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}$ = 2. The area cut off by the chord of contact on the region between the asymptotes is equal to-

(a) $\frac{ab}{2}$ (b) ab (c) 2ab (d) 4ab

9. If (a secθ, b tanθ) and (a secϕ, b tanϕ) are the ends of a focal chord of $\frac{x^2}{a^2}-\frac{y^2}{b^2}$ = 1, then tan $\frac{\theta}{2}\tan\frac{\phi}{2}$ equals to

(a) $\frac{e-1}{e+1}$ (b) $\frac{1-e}{1+e}$

(c) $\frac{1+e}{1-e}$ (d) $\frac{e+1}{e-1}$

10. Equation of hyperbola passing through origin and whose asymptotes are 3x + 4y = 5 and 4x + 3y = 5, is:

(a) $x^2 - y^2 = 1$

(b) $12x^2 + 12y^2 + 35xy - 15x - 15y = 0$

(c) $12x^2 + 12y^2 + 25xy - 35x - 35y = 0$

(d) $12x^2 + 12y^2 + 25xy - 25x - 25y = 0$

11. If x+y+k = 0 is a tangent to the parabola $x^2 = 4y$, then k =

(a) 1 (b) 2 (c) -1 (d) 4

12. The equation of the tangent to the parabola $y = (x - 3)^2$ parallel to the chord joining the points (3, 0) and (4, 1) is -

(a) $2x - 2y + 6 = 0$ (b) $2y - 2x + 6 = 0$

(c) $4y - 4x + 13 = 0$ (d) $4x + 4y = 13$

13. A line bisecting the ordinate PN of a point P $(at^2, 2at)$, $t > 0$, on the parabola $y^2 = 4ax$ is drawn parallel to the axis to meet the curve at Q. If NQ meets the tangent at the vertex at the point T, Then the coordinates of T are.

(a) (0, (4/3) at) (b) (0, 2at)

(c) $(1/4)\ at^2$, at) (d) (0, at)

14. If the tangent to the parabola $y^2 = ax$ makes an angle of 45^0 with x-axis, then the point of contact is

(a) $\left(\frac{a}{2}, \frac{a}{2}\right)$ (b) $\left(\frac{a}{4}, \frac{a}{4}\right)$

(c) $\left(\frac{a}{2}, \frac{a}{4}\right)$ (d) $\left(\frac{a}{4}, \frac{a}{2}\right)$

15. The length of the diameter of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$ perpendicular to the asymptotes of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ passing through the first and third quadrant is

(a) $\frac{100}{\sqrt{431}}$ (b) $\frac{150}{\sqrt{481}}$

(c) $\frac{25}{\sqrt{3}}$ (d) $11\sqrt{2}$

16. If a variable straight line $x \cos \alpha + y \sin \alpha = p$ which is chord of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2} = 1 (b>a)$ subtend a right angle at the centre of hyperbola, then it always touches a fixed circle whose radius is equal to -

(a) $\frac{ab}{a^2+b^2}$ (b) $\frac{ab}{\sqrt{b^2-a^2}}$

(c) $\frac{ab}{\sqrt{a^2+b^2}}$ (d) None

17. If P is a point on the rectangular hyperbola $x^2 - y^2 = a^2$, C is its centre and S, S′ are the two foci, then SP. S′ P =

(a) 2 (b) $(CP)^2$ (c) $(CS)^2$ (d) $(SS')^2$

18. The product of the lengths of perpendiculars drawn from any point on the hyperbola $x^2 - 2y^2 - 2 = 0$ to its asymptotes, is-

(a) ½ (b) 2/3 (c) 3/2 (d) 2

19. The slopes of the common tangent to the hyperbolas

$\frac{x^2}{9} - \frac{y^2}{16} = 1$ and $\frac{y^2}{9} - \frac{x^2}{16} = 1$ are-

(a) -2 (b) -1 (c) 2 (d) None

20. If the tangent and the normal to a rectangular hyperbola at a point cut off intercepts a_1, a_2 on one axis and b_1, b_2 the other axis then $a_1a_2 + b_1b_2$ is equal to-

(a) -1 (b) 0 (c) 1 (d) $\sqrt{2}$